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of Glasgow

# Big Data: Systems, Programming and Management

COMPSCI4064 (H) and COMPSCI5088 (M)

Dr. Richard McCreddie

**WORLD  
CHANGING  
GLASGOW**

**A WORLD  
TOP 100  
UNIVERSITY**



## What is this course about?

- The Big Data course is designed to teach you how to **identify big data problems** and **provide you the tools to solve those problems**
- Broadly it covers three main aspects
  - **Theory** of what Big Data is and why it's a problem
  - Technologies to enable **processing** of Big Data
  - Technologies to enable the **storage and access** of Big Data





# Intended Learning Outcomes (ILOs)

## Design, develop and evaluate

Design, develop and evaluate programs to access big data repositories in a massively parallel manner;

## Discuss and contrast

Discuss and contrast the internals of the design and implementation of current cloud data storage and processing systems;

## Identify and analyse

Identify and analyse issues related to the scalability and efficiency challenges of processing complex queries/algorithms against big data systems, and develop and assess ways of addressing said challenges;

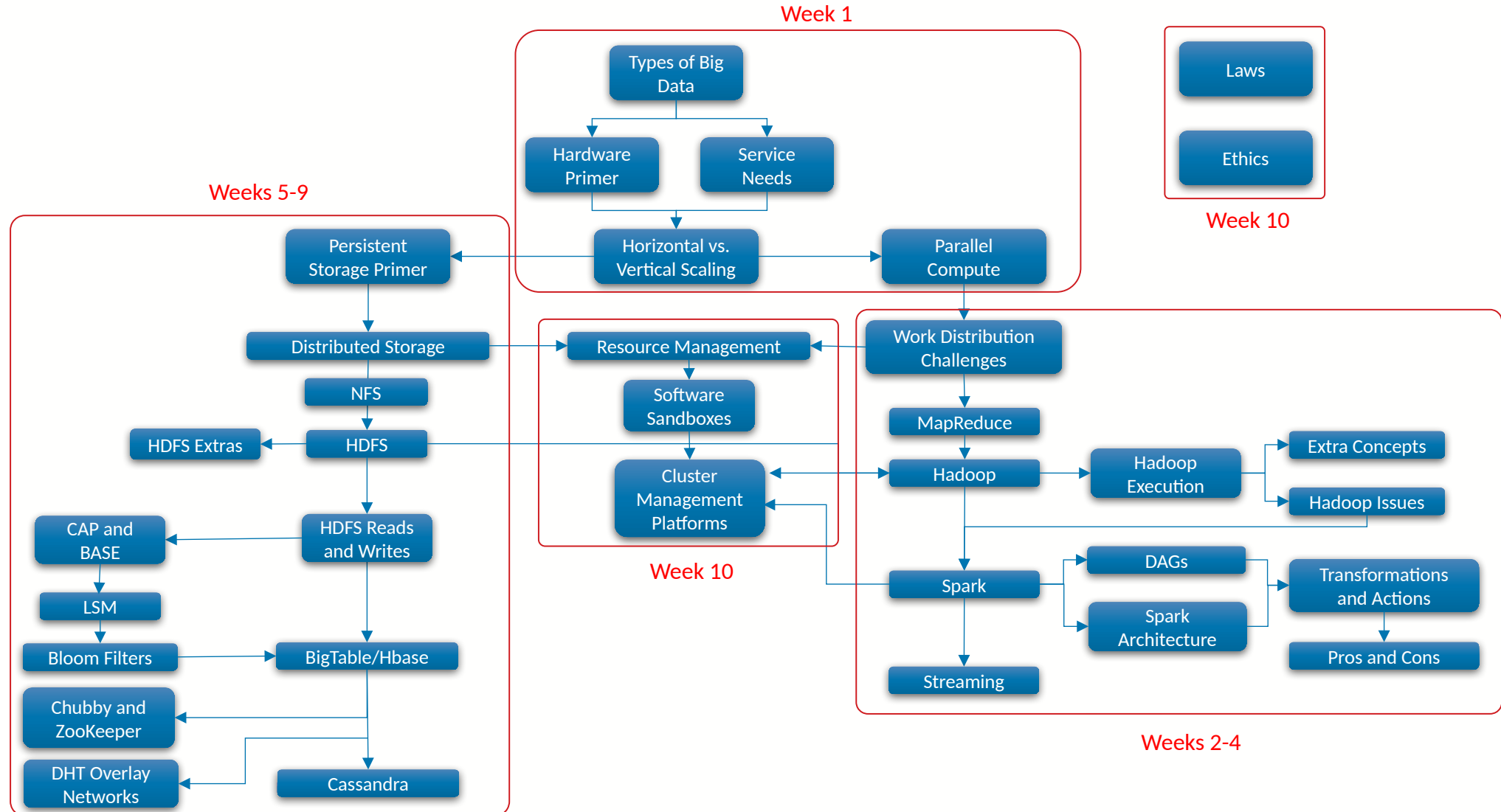
## Explain and appraise

Explain and appraise state-of-the-art research in the field of big data systems.

## Demonstrate

Demonstrate that they have mastered the required background knowledge to pursue graduate studies in the fields of cloud systems and big data.

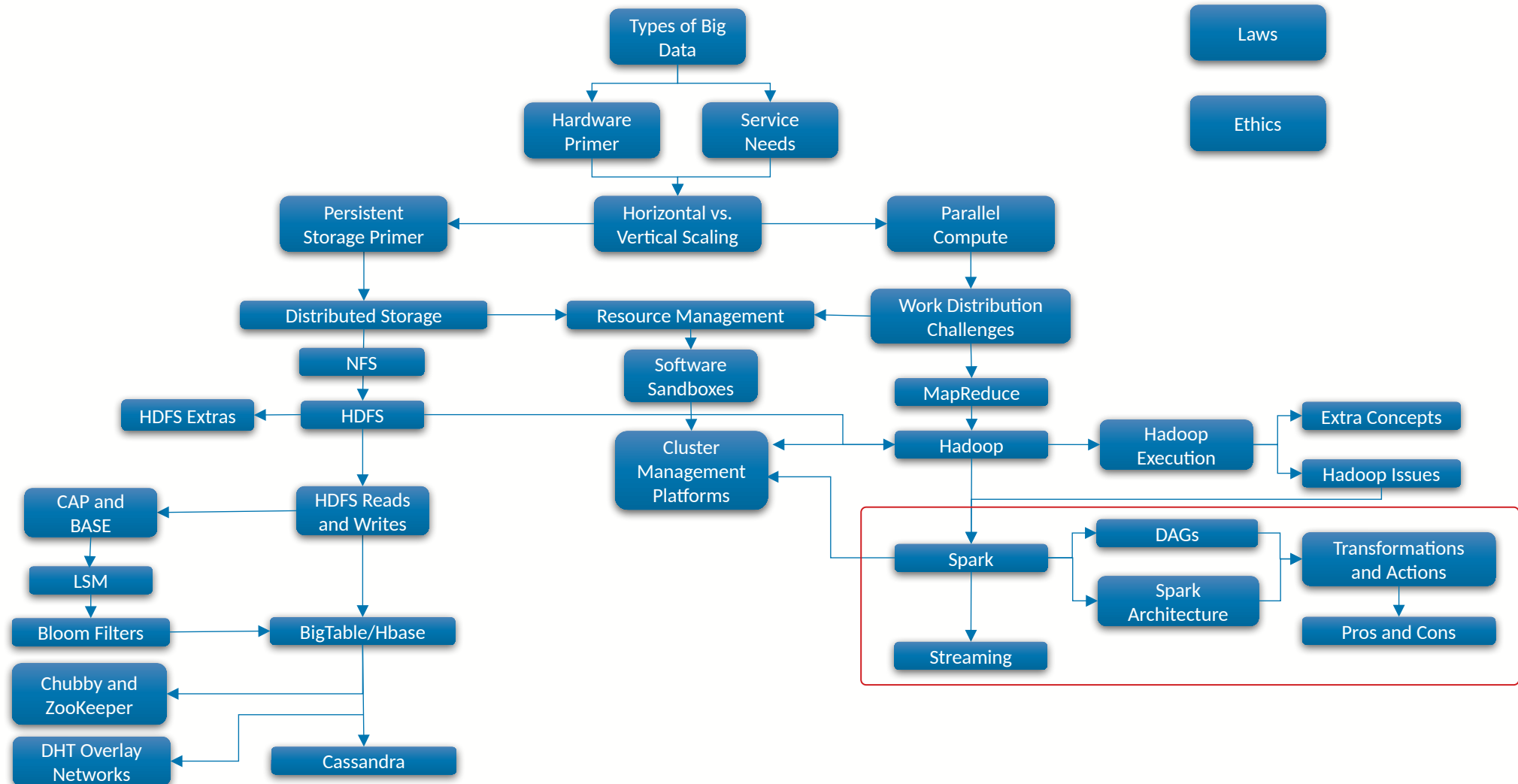
# (Examinable) Topics Covered







# Hands on Skills





# Course Content

- This course provides a mix of live and online content
  - Live
    - **2 hour lecture per week** (video available)
      - Monday 1 pm to 3pm
      - Boyd Orr Building: Lecture Theater 1 / Room 203
    - **1 hour lab per week** (starting in week 3)
      - Wednesday 12 noon-1pm (BO 1028)
      - Wednesday 3pm-4pm (BO 1028)
      - Wednesday 4pm-5pm (BO 1028)
  - Online
    - Bite-sized Topic Videos
    - Apache Spark Tutorials
    - Background Reading



# Course Change Log

## 2024-25

- New assessed exercise was added and moved from team based to individual assessment
- [In Progress...] Adding optional Apache Flink tutorial content

## 2023-24

- Updated exam formatting with more detailed answer guidance blocks
- Added Batch vs. Streaming Content (W4-06)
- Recorded additional bite-sized lecture recordings

## 2022-23

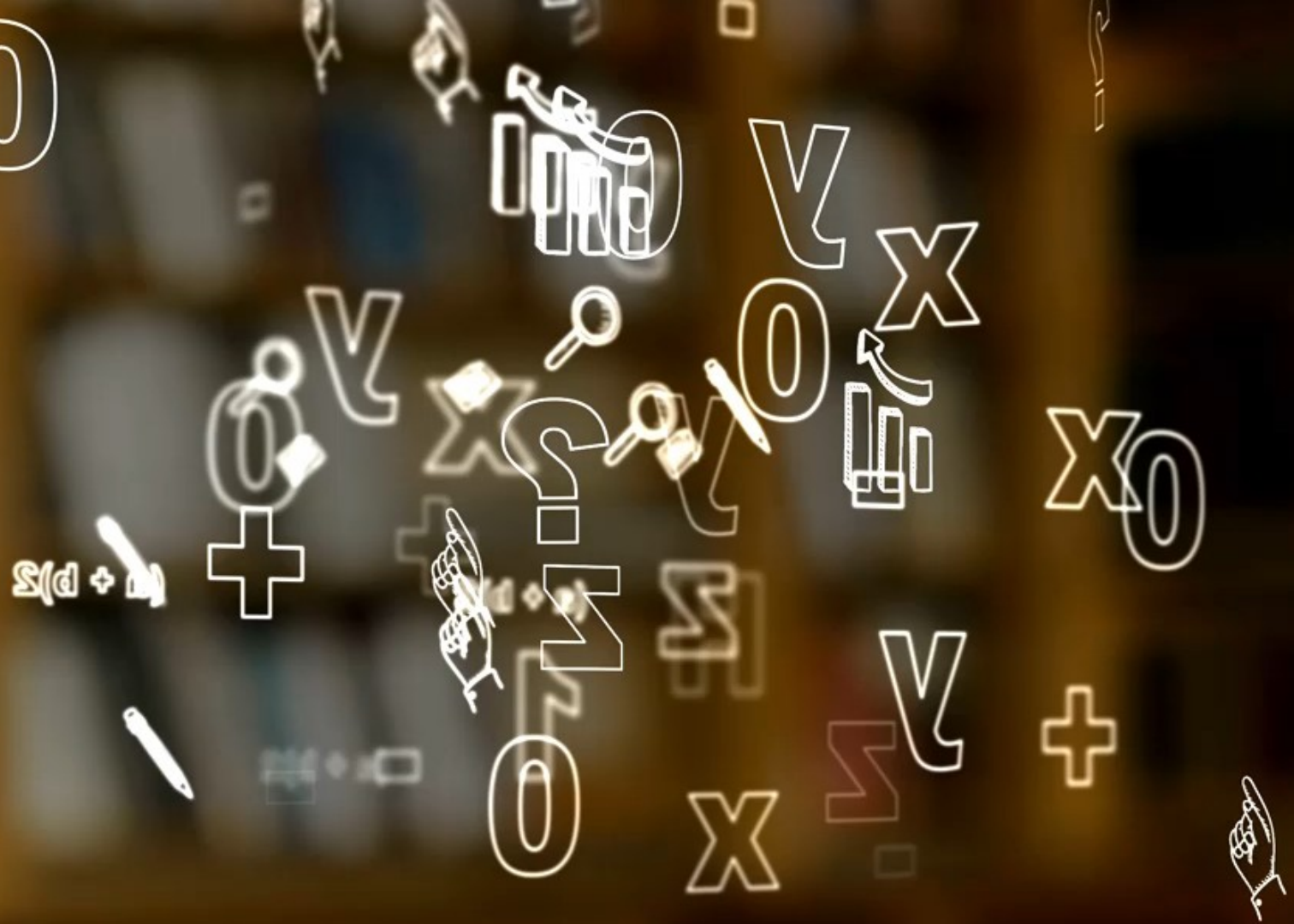
- Lecture material (slides/videos) were completely replaced
- Added a third hour of lab time each week
- Online Quizzes removed (due to time constraints)

## 2021-22

- Added Apache Spark tutorials and video material
- Added new Apache Spark-based assessed exercise, removed Hadoop-based assessed exercise

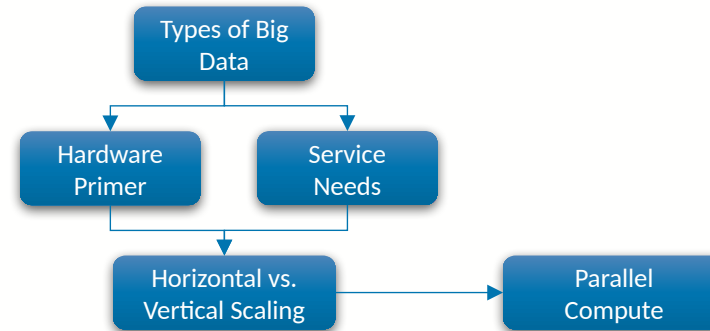


# Lecture Structure



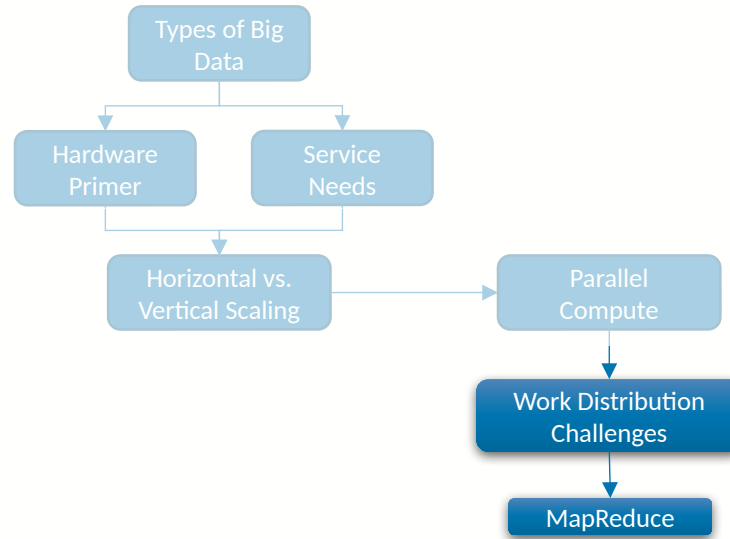


# Week 1



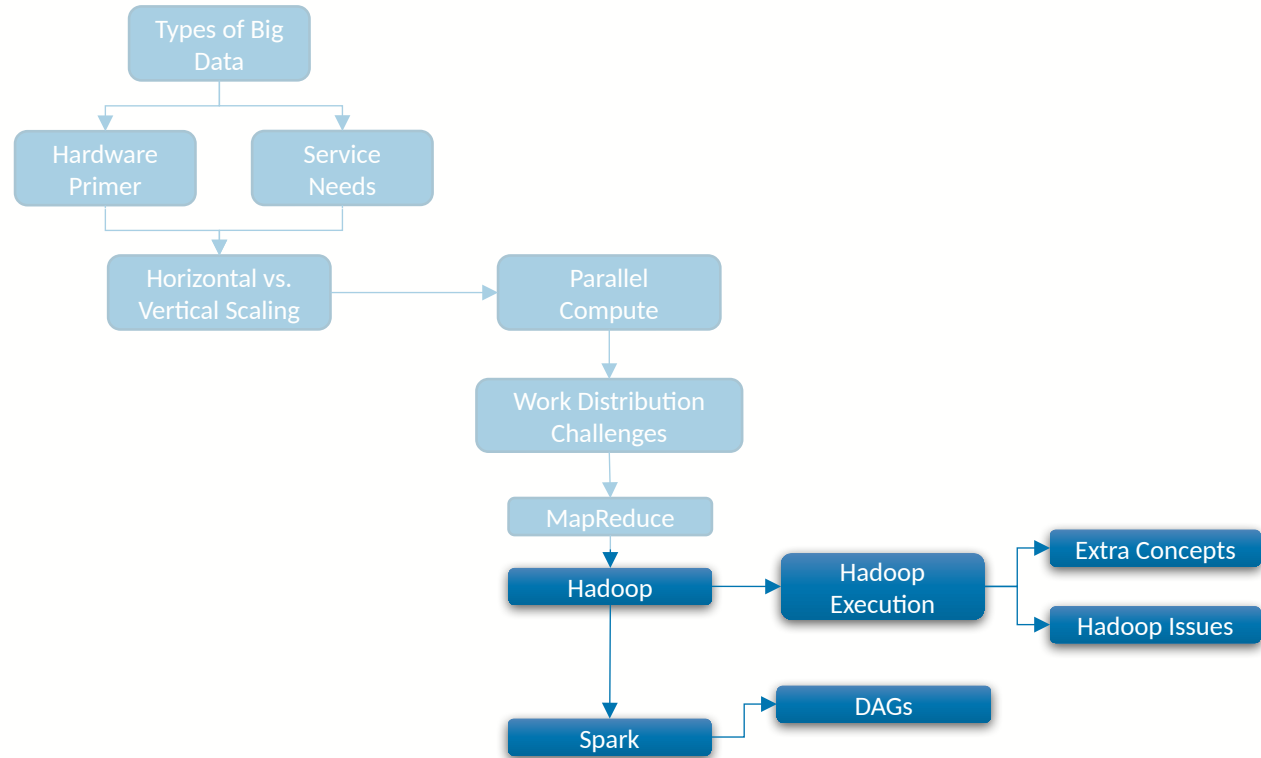


## Week 2



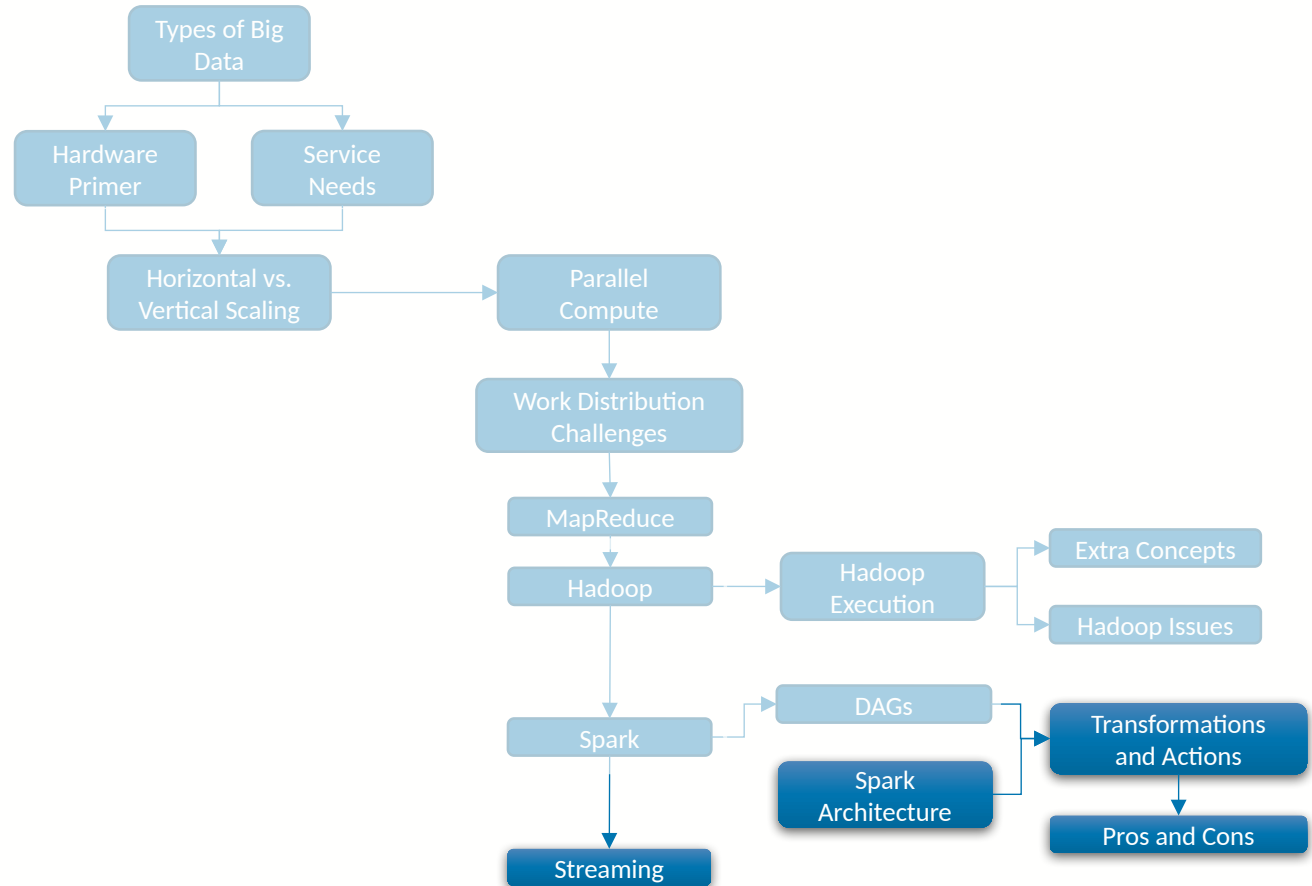


## Week 3





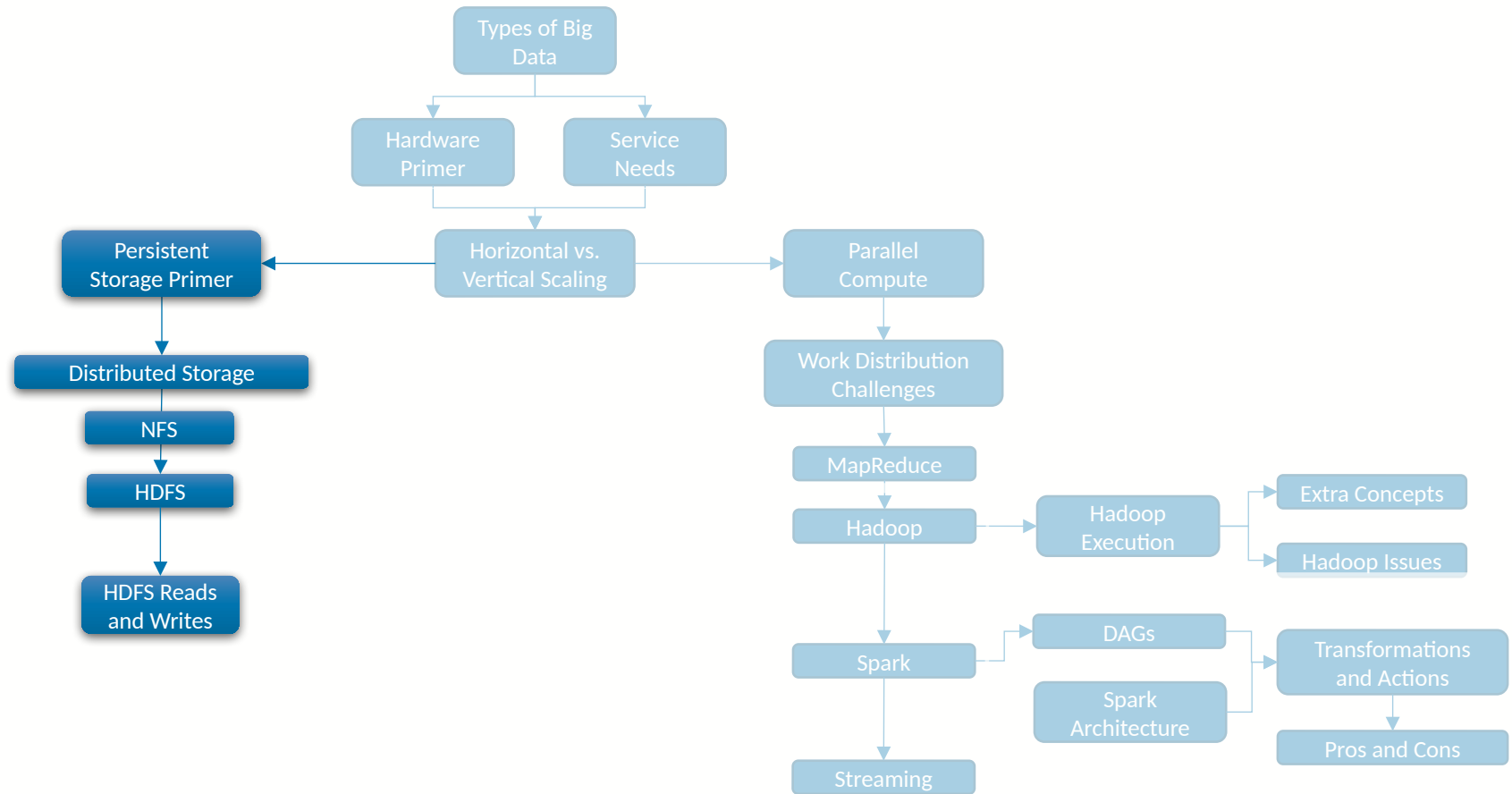
## Week 4





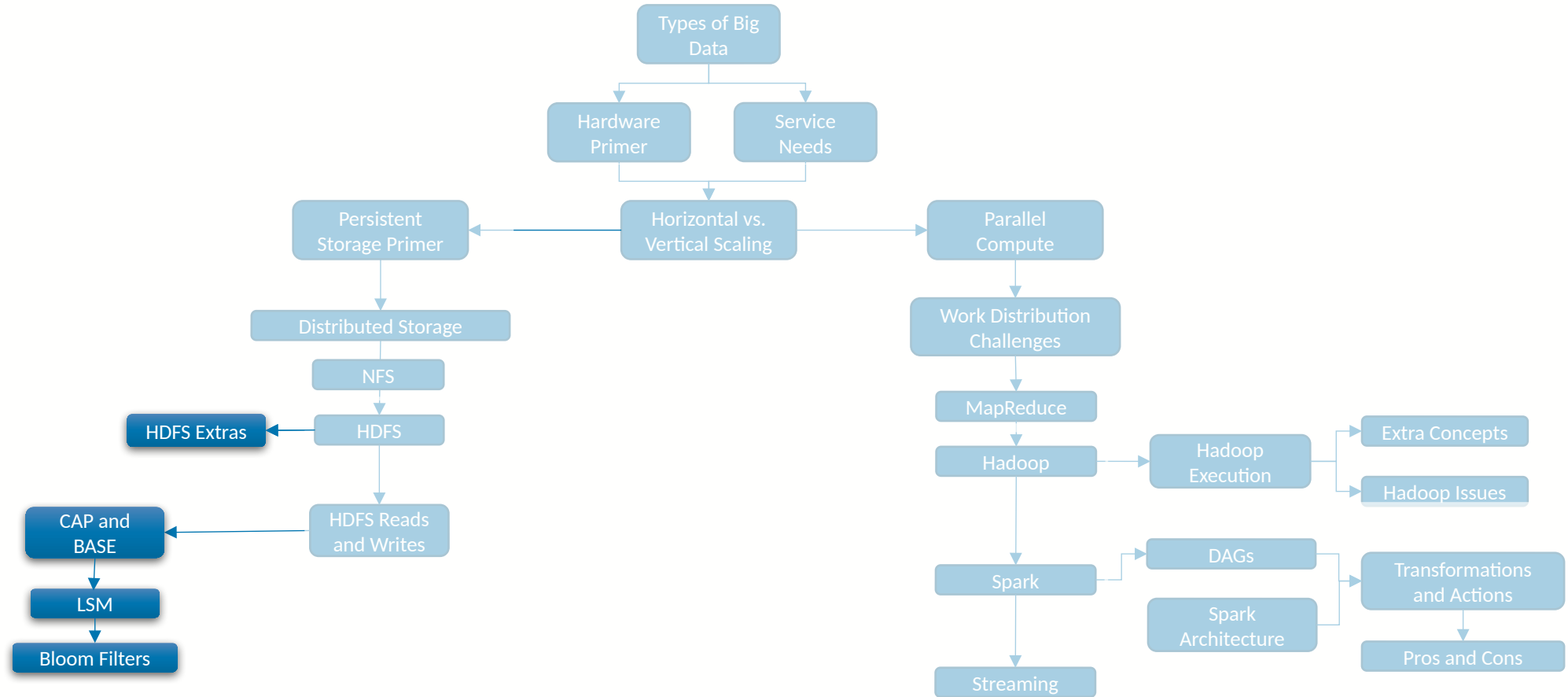


## Week 5



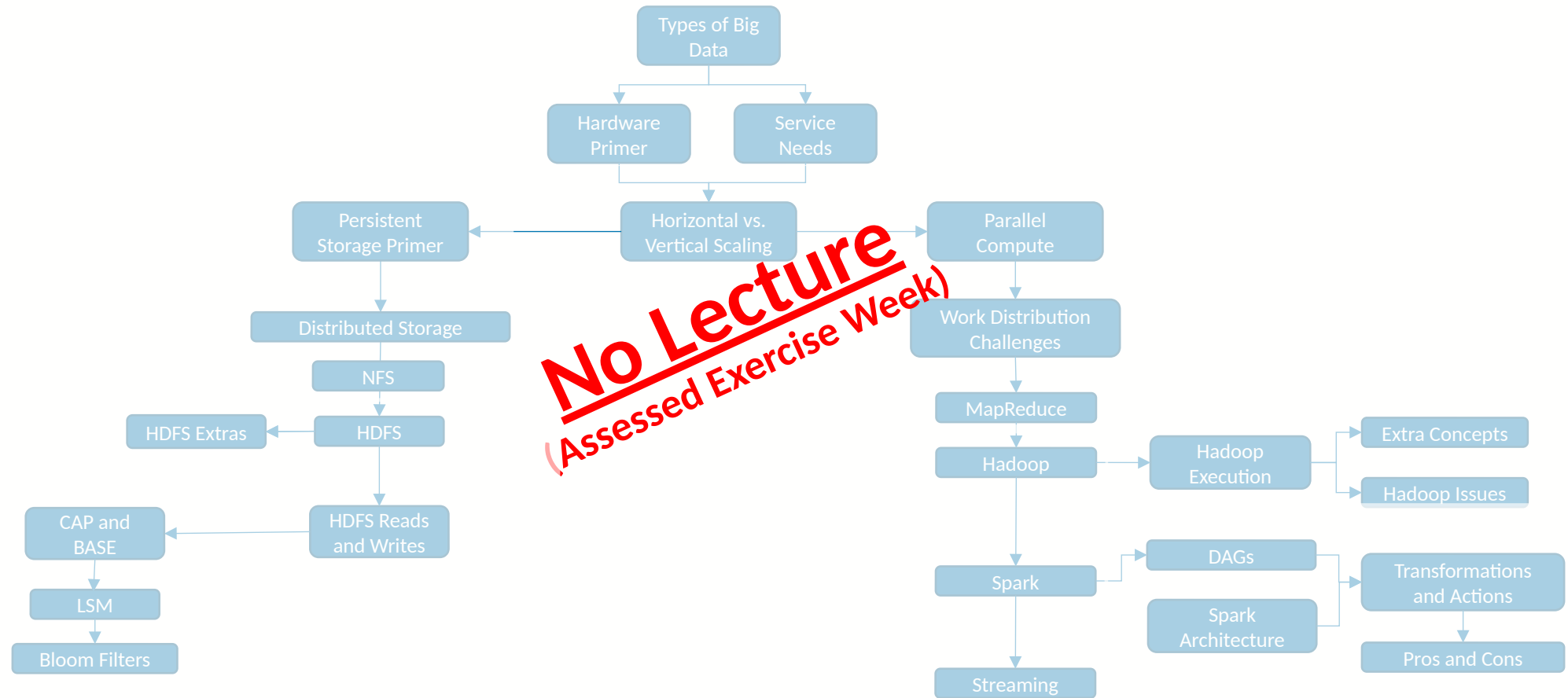


## Week 6



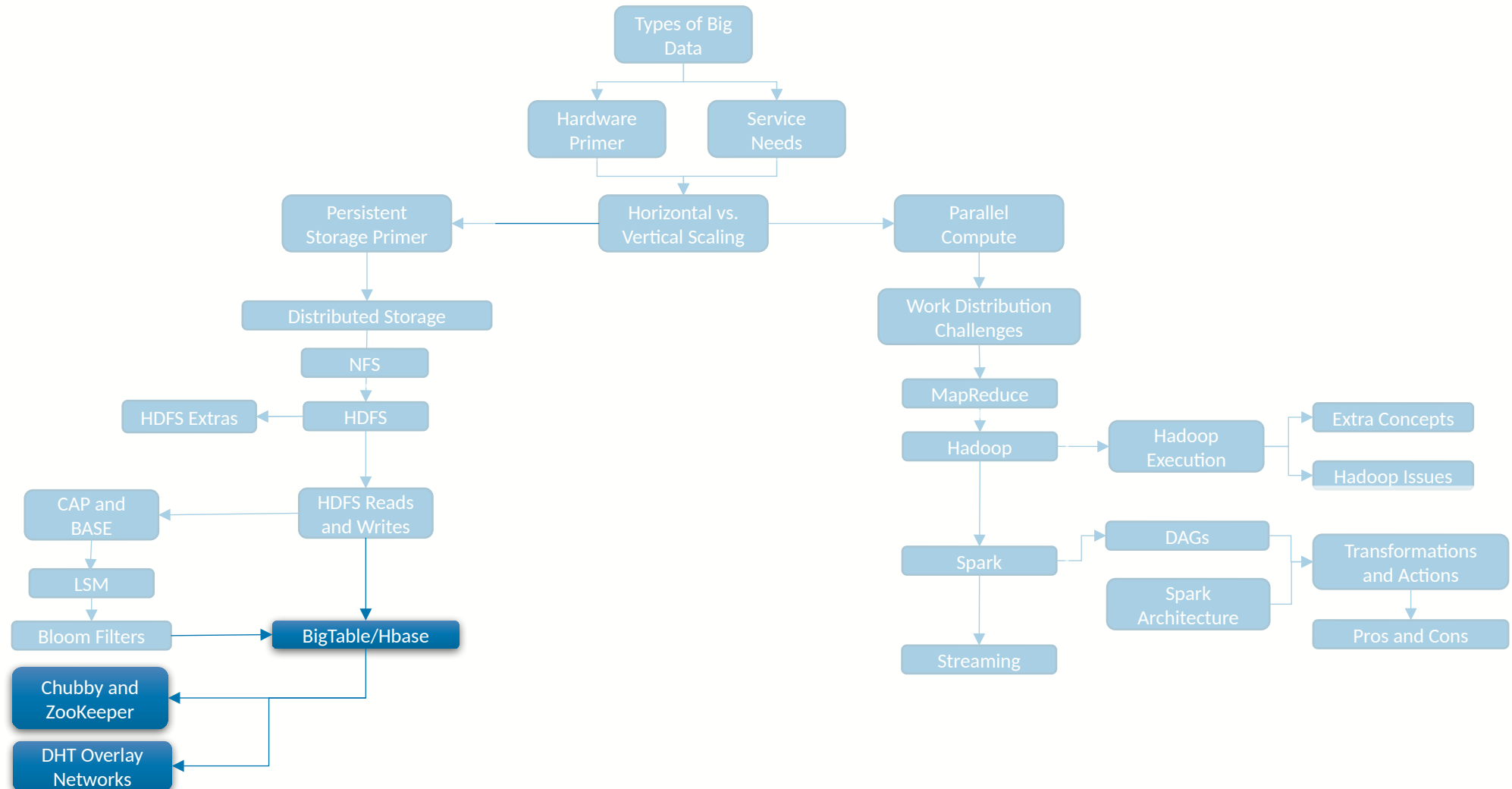


## Week 7



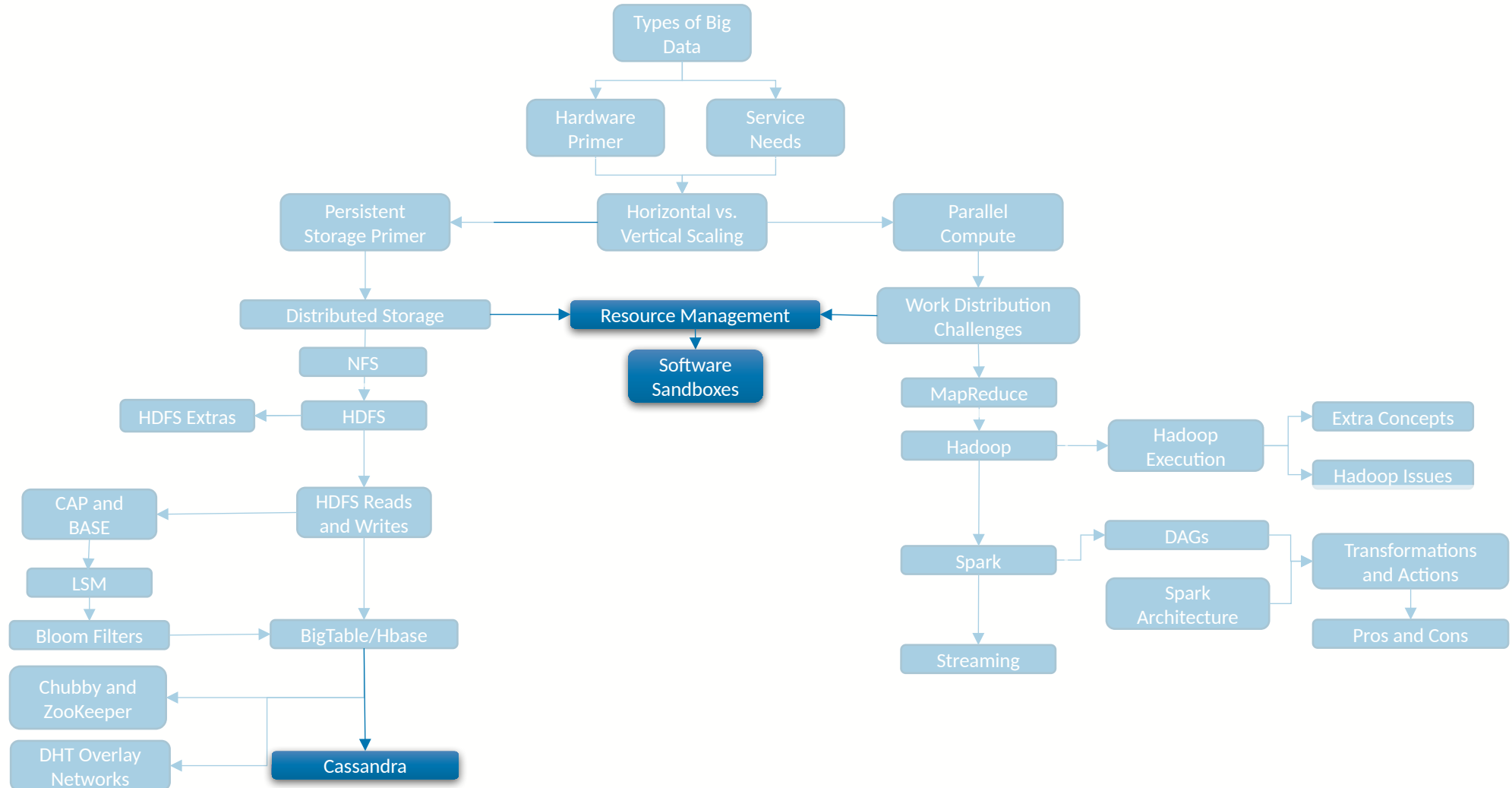


## Week 8



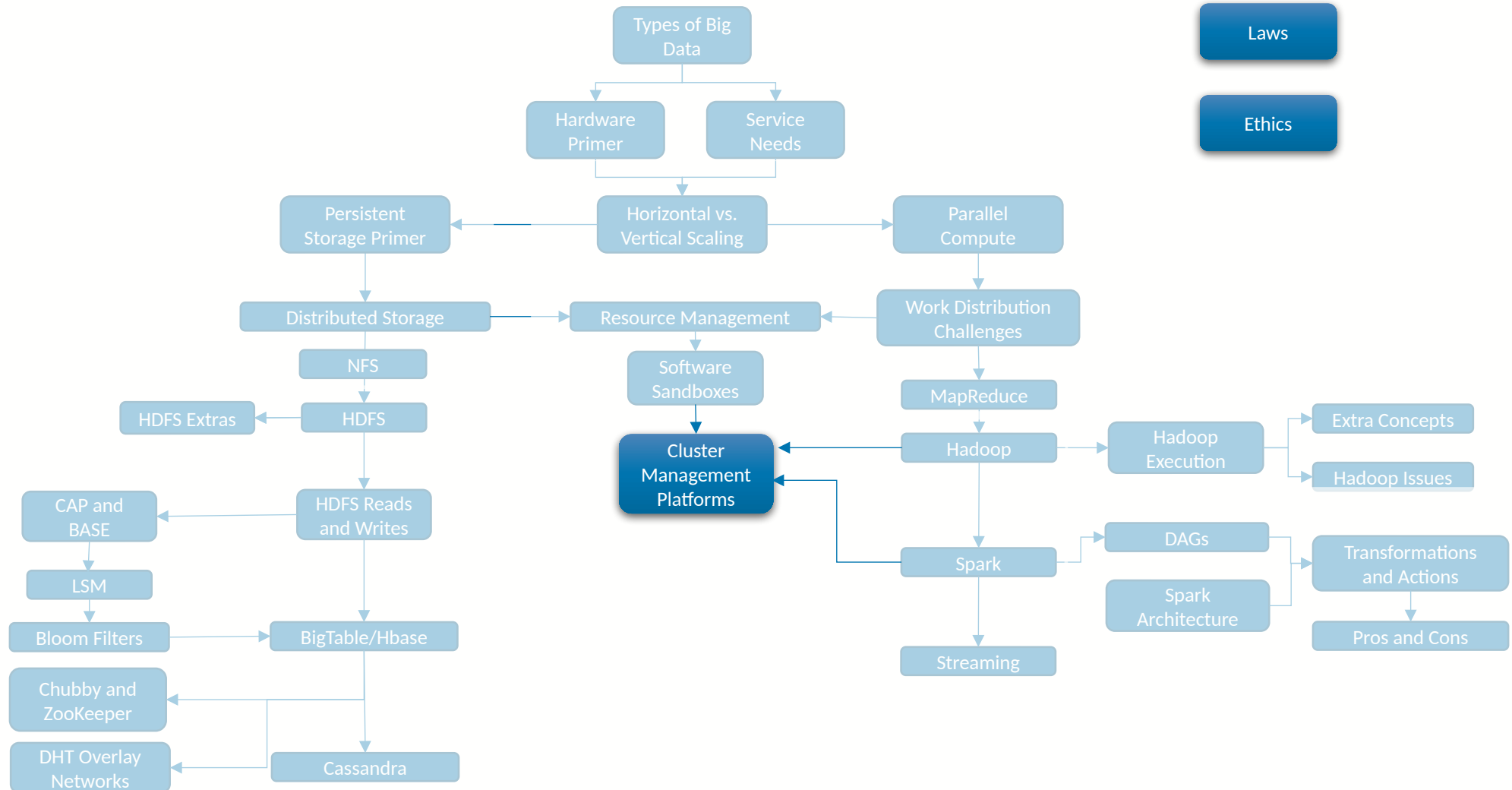


## Week 9





# Week 10



Laws

Ethics





# Week 11



# Assessment





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## Assessed Exercise



- Building an Apache Spark Application
  - Java Programming
- Hand-out: Week 3
- Deadline: **March 3<sup>rd</sup>**
- **25%** of your course mark



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## Exam



- Written Exam
  - H Exam: 2 hours, 60 marks
  - M Exam 2 hours, 60 marks
    - (Some questions are different for the M exam)
- Explain, Describe, Discuss-type questions
- 75% of your course mark



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## Computer Labs





## Why Labs?

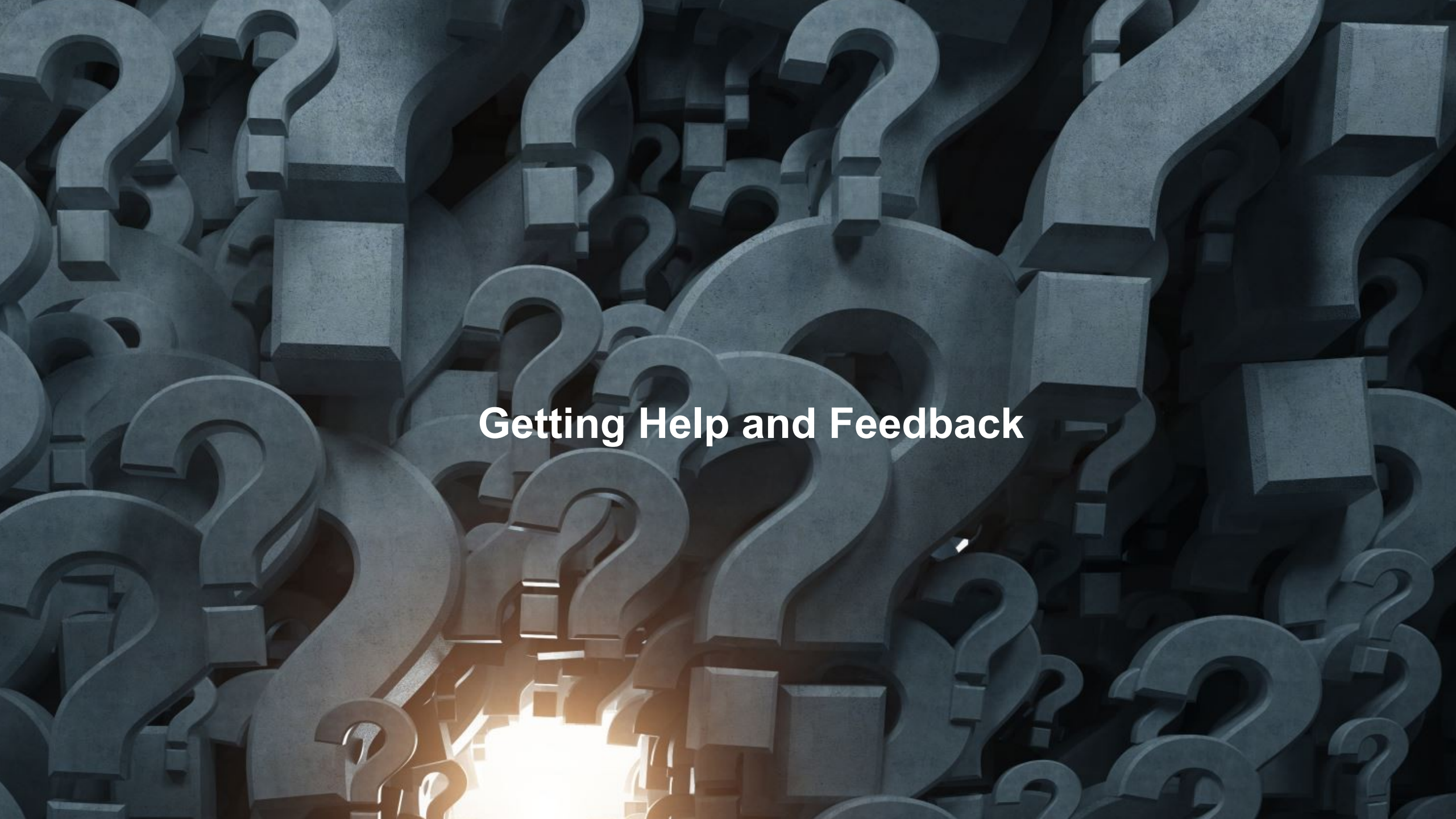
- The course has a hands-on component where you will be learning to program using Apache Spark, which also prepares you for the assessed exercise
- We have **lab sessions** on **Wednesday afternoon** each week to help you
  - Labs start in Week 3 (Jan 29<sup>th</sup>), which coincides with when we introduce Spark in the lectures
  - I will be there as well as tutors



## Lab Capacity

- The lab has a capacity of ~80 students at any one time
- This class is around 200 students
- We have **three 1-hour lab sessions in Wednesday afternoon**, such that everyone has an opportunity to turn up
  - The labs are your best opportunity to **ask questions and get help!**



The image features a dense, overlapping field of three-dimensional question marks. The marks are rendered in a dark, metallic blue-grey color with a matte finish. They vary in size and are arranged in a way that creates a sense of depth and complexity. At the bottom center of the image, there is a bright, glowing light source that casts a warm, orange-yellow glow upwards, illuminating the question marks in the foreground and creating a strong contrast with the darker background. The overall composition suggests a state of confusion, inquiry, or a search for answers.

**Getting Help and Feedback**



## Places to Get Help and Live Feedback

- Lectures

- The lectures are divided into content blocks, and we will have short gaps between blocks, feel free to ask me questions during those gaps

- Labs

- Both myself and tutors will be available during the lab sessions, come and ask us questions!

- Office Hour

- My open-door office hour each week is Friday 12 noon to 1pm (Sir Alwyn Williams Building, room 304), you don't need an appointment, just drop by

- Other contact avenues

- **Open Forum:** There is an open forum on the Moodle page, feel free to post questions in there and I'll check in and answer them 1-2 times week
- **Mail/Teams:** For sensitive topics you can email or message me on Teams, my email is [richard.mccreadie@glasgow.ac.uk](mailto:richard.mccreadie@glasgow.ac.uk)



# Assessment Feedback

- **Assessed Exercise**

- If working on this, the best place to check if you are on the right path is to ask myself or the tutors in the Labs
- After marking you will get a detailed mark breakdown for your submission, and I will also provide a video talking through my solution

- **Exam**

- A practice exam will be provided and there are also past exams with marking schemes and answers available on the Moodle page
- After the exam, the solution will be posted to Moodle.





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